Ordinary Hausdorff density is the tangent isodiametric volume

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Status. Full affirmative solution, high confidence. The conjecture in arXiv:1005.0540 follows from a uniform blow-up lemma in which each competing set is rescaled by its *own* diameter. The same lemma also proves continuity of the density and answers all three ordinary Hausdorff-measure questions later isolated in arXiv:1702.00241.

1 Source conjecture

Let $(M, \mathcal{D}, g)$ be an equiregular sub-Riemannian manifold of Hausdorff dimension Q, let μ be a smooth volume, and let $\mathcal{H}^Q$ be ordinary (not spherical) Hausdorff measure, with covering cost $(\operatorname{diam} A)^Q$. Agrachev, Barilari, and Boscain conjecture that the density in $d\mu = f_{\mu\mathcal{H}}\, d\mathcal{H}^Q$ is the largest induced tangent volume of a set of diameter one [1]:

> the volume of the unit sub-Riemannian ball of the nilpotent approximation of the sub-Riemannian manifold, that can be explicitly described in a certain number of cases (see Theorem 1 below). On the other hand nothing is known on how to compute $f_{\mu\mathcal{H}}$. We conjecture that $f_{\mu\mathcal{H}}$ is given by the μ-volume of certain isodiametric sets, i.e. the maximum of the μ-volume among all sets of diameter 1 in the nilpotent approximation (see [32, 40] and reference therein for a discussion on isodiametric sets). This quantity is not very natural in sub-Riemannian geometry and is extremely difficult to compute.
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> Our interests in studying $f_{\mu\mathcal{H}}$ comes from the following question:

At $p \in M$, write $(G_p, \widehat{d}_p)$ for the nilpotent approximation and $\widehat{\mu}_p$ for the induced Haar volume, and set

$$C(p) := \sup\{\widehat{\mu}_p(A) : A \subset G_p,\ \operatorname{diam}_{\widehat{d}_p} A = 1\}. \tag{1}$$

The supremum can equivalently be taken over compact sets and is attained by the existence theorem for isodiametric sets [3].

Theorem 1. *For every regular point p,*

$$\frac{d\mu}{d\mathcal{H}^Q}(p) = C(p).$$

Here the left side is its canonical Federer-density representative; in particular $d\mu = C(\cdot)\, d\mathcal{H}^Q$. Moreover C is continuous on every equiregular component.

Consequently the tangent formula conjectured in the source is true. In the notation of Ghezzi–Jean [2], the local isodiametric constants satisfy $\mathscr{I}(p, \epsilon) \to \mathscr{I}_p$. The function $p \mapsto \mathscr{I}_p$ and the density $d\mathcal{H}^Q/d\mu = 1/C(p)$ are continuous, and the ordinary Hausdorff metric-measure blow-ups converge to $(G_p, \widehat{d}_p, \mathcal{H}^Q_{\widehat{d}_p}, e)$.

2 Proof intuition

Federer differentiation expresses $d\mu/d\mathcal{H}^Q(p)$ as a supremum over all sufficiently small closed sets containing p. Arbitrary shapes are not the real difficulty; the scale is. If all competitors are enlarged by the same outer cutoff ϵ, a set much smaller than ϵ sees a useless additive metric error. Instead, rescale each set by its own diameter t. Uniform pairwise convergence of the rescaled sub-Riemannian distance then makes its tangent diameter $1 + o(1)$, uniformly over every possible shape. Smoothness of the volume form gives an equally uniform volume comparison. The tangent isodiametric inequality does the rest.

3 The uniform-supremum lemma

Choose privileged coordinates x centered at p, write δ_t for their anisotropic dilations, and define

$$J(p,\epsilon) := \sup\left\{ \frac{\mu(S)}{(\operatorname{diam}_d S)^Q} : S \subset M \text{ closed},\ p \in S, 0 < \operatorname{diam}_d S < \epsilon \right\}. \tag{2}$$

Lemma 1. $\lim\limits_{\epsilon \downarrow 0} J(p,\epsilon) = C(p)$.

Proof. The precise metric blow-up used below is the standard uniform pairwise convergence, recorded explicitly in the proof of Ghezzi–Jean, Proposition 3.3 [2]:

$$a(t) := \sup_{u,v \in \overline{B}_d(p,t)} \left| \frac{d(u,v)}{t} - \widehat{d}_p\big(\delta_{1/t}x(u), \delta_{1/t}x(v)\big) \right| \longrightarrow 0. \tag{3}$$

Take a competitor S in (2) and put $t = \operatorname{diam}_d S$. Since $p \in S$, it is contained in $\overline{B}_d(p,t)$. The rescaled set $A_t = \delta_{1/t}x(S)$ lies in one fixed compact set K by the ball–box theorem. Equation (3) gives, uniformly over all such S,

$$1 - a(t) \le \operatorname{diam}_{\widehat{d}_p} A_t \le 1 + a(t). \tag{4}$$

Write $d\mu = g(z)\,dz$ in these coordinates. The anisotropic Jacobian of δ_t is t^Q, while $d\widehat{\mu}_p = g(0)\,dz$. Hence

$$t^{-Q}\mu(S) = \int_{A_t} g(\delta_t z)\,dz. \tag{5}$$

If $b(t) = \sup_{z \in K} |g(\delta_t z) - g(0)|$, then $b(t) \to 0$ and

$$\left| t^{-Q}\mu(S) - \widehat{\mu}_p(A_t) \right| \le b(t)|K|.$$

By homogeneity of (1), $\widehat{\mu}_p(A) \le C(p)(\operatorname{diam}_{\widehat{d}_p} A)^Q$. Together with (4), this proves

$$\frac{\mu(S)}{t^Q} \le C(p)(1 + a(t))^Q + b(t)|K|,$$

uniformly for all competitors, and therefore $\limsup_{\epsilon \downarrow 0} J(p,\epsilon) \le C(p)$.

Conversely, fix $\eta > 0$ and choose a compact $A \subset G_p$ of diameter one with $\widehat{\mu}_p(A) > C(p) - \eta$. A left translation preserves both quantities, so assume $e \in A$. For small t, let $S_t = x^{-1}(\delta_t A)$. Uniform metric convergence and (5) give

$$\frac{\operatorname{diam}_d S_t}{t} \to 1, \qquad \frac{\mu(S_t)}{t^Q} \to \widehat{\mu}_p(A).$$

Thus $\liminf J(p,\epsilon) \ge C(p) - \eta$. Letting $\eta \downarrow 0$ finishes the proof. $\square$

4 Density and continuity

Ghezzi–Jean record the Federer formula [2]

$$\frac{d\mu}{d\mathcal{H}^Q}(p) = \lim_{\epsilon\downarrow 0} J(p,\epsilon). \tag{6}$$

Theorem 1's density identity follows immediately from Lemma 1. This direct route also fixes the normalization: in Euclidean space it reads $d\mathcal{L}^n/d\mathcal{H}^n = \omega_n/2^n$, the volume of a diameter-one Euclidean ball.

It remains to prove the asserted continuity. Around p, choose privileged coordinates smoothly in q and identify all G_q with $\mathbb{R}^n$. The tangent radial distances vary uniformly on compact sets [1]; since their group laws also vary smoothly, the pairwise distances $\widehat{d}_q$ do likewise. The induced Haar densities vary uniformly on compact sets as well.

Let $q_j \to p$. For each j, translate a compact diameter-one near-maximizer A_j for $C(q_j)$ so that it contains the identity. Uniform ball–box estimates place every A_j in one compact K. Uniform convergence of distance and density gives

$$\operatorname{diam}_{\widehat{d}_p} A_j = 1 + o(1), \qquad \widehat{\mu}_p(A_j) = \widehat{\mu}_{q_j}(A_j) + o(1),$$

and hence $\limsup_j C(q_j) \le C(p)$. For the reverse inequality, use one compact near-maximizer for $C(p)$ as a competitor in every tangent structure at q_j. Its diameter and volume converge, so

$$\liminf_j C(q_j) \ge C(p).$$

This proves continuity.

Finally, the spherical density formula of [1] and Rigot's identity $\mathcal{S}^Q = \mathscr{I}_p\mathcal{H}^Q$ on G_p give the independent check

$$\mathscr{I}_p = \frac{2^Q C(p)}{\widehat{\mu}_p(\widehat{B}_p)}, \qquad \frac{d\mathcal{H}^Q}{d\mu}(p) = \frac{2^Q}{\mathscr{I}_p\widehat{\mu}_p(\widehat{B}_p)} = \frac{1}{C(p)}.$$

Continuity of $C(p)$ and of tangent ball volume proves continuity of $\mathscr{I}_p$. Proposition 3.10 of [2] then gives the stated measured Gromov–Hausdorff convergence.

5 Scope and verification

The theorem concerns regular/equiregular points. It does not claim an ordinary-Hausdorff tangent formula at singular points, where the homogeneous dimension and volume behavior can change. No classification or explicit shape of tangent isodiametric sets is needed.

The proof was audited at four levels: the cutoff-versus-own-diameter scale issue; the uniform Jacobian bound for arbitrary varying sets; the direct Federer normalization; and continuity under varying tangent structures. The source, the later equivalent formulation, and Rigot's tangent normalization were all checked in primary sources. Bounded searches through 13 August 2026 found no later proof of these exact questions.

References

[1] A. Agrachev, D. Barilari, and U. Boscain, *On the Hausdorff volume in sub-Riemannian geometry*, Calc. Var. PDE 43 (2012), 355–388; arXiv:1005.0540.

[2] R. Ghezzi and F. Jean, *On measures in sub-Riemannian geometry*, Sém. Théor. Spectr. Géom. 33 (2015–2016), 17–46; arXiv:1702.00241, especially Sections 3.2–3.3.

[3] S. Rigot, *Isodiametric inequality in Carnot groups*, Ann. Acad. Sci. Fenn. Math. 36 (2011), 245–260; arXiv:1004.1369.